

Department of Electrical, Electronic and Computer Engineering

Assessment ID: 2021EBN111A01

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Electricity and Electronics EBN 111

19 March 2021

Marks:	10	Total number of pages:	6
Submission Deadline:	23 March 2021 (23:59)	Open/Closed book:	Open

Important instructions

1. Write your answers on the Excel spreadsheet found on the [AMS system](#). The spreadsheet contains unique parameter values for each student. Therefore, you must download the Excel spreadsheet from your own AMS account.
2. Follow the instructions on the AMS spreadsheet carefully.
3. Final answers must be written as real numbers and not as fractions. (Use at least 5 significant figures to write irrational numbers.)
4. Answers must include units where applicable. (Refer to the [EECE Rules for Electronic Assessments of tests and examinations](#) for guidelines on how to enter units. Also refer to the example list of units on the Excel spreadsheet.)

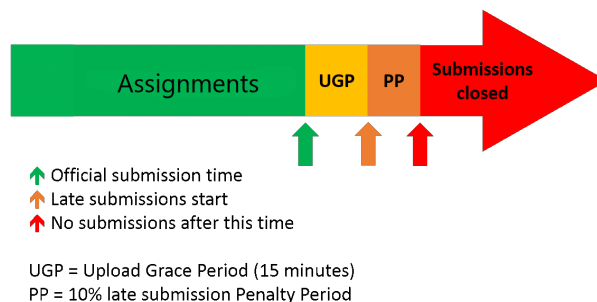
Lecturers: Dr D. le Roux, Dr S. Smith, Dr X. Ye

Assistant Lecturers: M. Maritz, J. Masela, J. Thiselton, M. Trivedi, D. Wepener

Online assessment submission rules and time-line

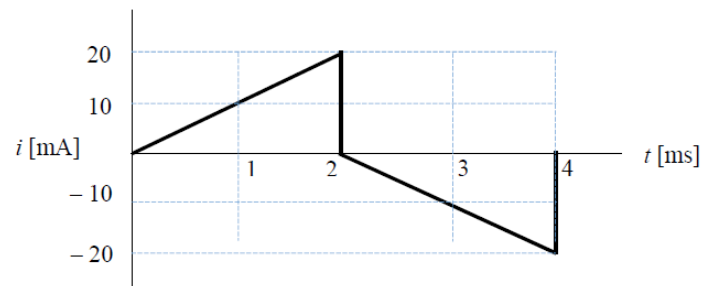
The following rules and time-line apply for all Assignments for EBN111/122 as illustrated by the figure below:

1. The start of the green section in the figure indicates the release of an assignment. The release of the assignment will be announced on ClickUP.
2. The official submission deadline for the assignment is indicated by the green arrow in the figure, which corresponds to the due dates specified in EBN Lecture Schedule.
3. You must submit your Excel spreadsheet with your answers to each assignment on the AMS before the due dates.
4. If you experience any technical issues with respect to uploading your assignment answer spreadsheet on the AMS, you can request assistance via **e-mail** ebn2021.ams@gmail.com. Note, except for extraordinary cases, **do not** submit your answers via email.
5. The submission deadline is followed by an Upload Grace Period of 15 minutes during which you must submit your assignment answers on the AMS even if not finished with the assignment. No email submissions will be accepted during this period.
6. The Upload Grace Period is followed by a Penalty Period of 10 minutes. If you submit during this period, a 10% penalty will be applied to your assignment mark.
7. No late submissions is accepted after the Penalty Period.



Question 1 [2]

The current flowing through a $[R] \Omega$ resistor is shown in the graph below. Calculate the charge Q through the resistor between $t = [t]$ ms and $t = 4$ ms.



01 Q [2]

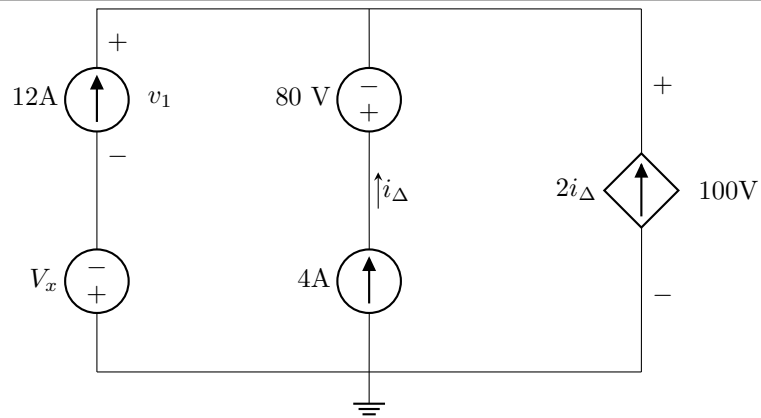
Question 2 [2]	
To move $-[Q]$ nC in $[X]$ μ s through an element requires $[E]$ mJ. Determine the voltage V across the element, assuming a constant current.	
02	V [2]

Question 3 [2]	
The total energy used by a light bulb in $[t]$ hours is $[E]$ kJ. The voltage across the bulb is $[v]$ V. Determine the current I through the bulb.	
03	I [2]

Questions 4 to 5 [2]	
The voltage across an element is $V(t) = \frac{di}{dt}$ V and the current through the element is $i(t) = [X]e^{-[Y]t}$ mA. The power $p(t) = Ae^{Bt}$ (in μW) associated with the element as a function of time. Determine A and B.	
04	A [1]
05	B [1]

Question 6 [2]

Calculate the power P delivered by the voltage source $V_x = [V]$.



06 P [2]

Grand Total : [10]